

Development of pandas AI framework in the domain of nuclear industry

A substantial large database with open AI domain will be developed by integrating pandas dataframe with open AI and prompt engineering technology.

Research Objective

A nuclear power plant consists of a huge amount of information, which if maintained and retrieved properly, can be very useful for us.

There can be various kinds of nuclear reactors like boiling water reactor, pressurised heavy water reactor, Pressurised water reactor and fast breeder reactor. All these different reactors have their own characteristics and specifications.

We can maintain a database consisting of the informations like the fuel used, the shape of the fuel, the moderator used, the coolant, the primary heat exchanger, condenser, grid or the generator required for each of these reactors.

1)In order to design a nuclear reactor, the designer need some specification where this database can help him to gain prior information about the existing nuclear reactors and their designs.

2)Next, we can also design a database regarding the design safety of each of these reactors.

3)Next, we can also design a database regarding the operational safety of each of these reactors.

4)Next comes the environmental impacts or environmental safety features.

As a nuclear reactor generates both gaseous and liquid waste, gaseous wastes are normally discharged through a stack, the design specification or operational safety of the stack is also necessary.

After creating all these databases, and incorporating AI into our database, we can create an interactive database. We can also implement algorithms to avoid data redundancy in our database. For example, if a designer builds a reactor, along with the specifications of the reactor design, with the help of prompt he can interactively search for the operational safety and environmental concerns associated to that particular reactor. Thus, such smart database can help in manipulation and visualisation of data in an overall basis, helping a lot not only in the field of reactor design but also nuclear safety which is a major concern in today's world.